Real-Time Social Media Misinformation Detection Using Natural Language Processing with TF-IDF and Text Classification

A CAPSTONE PROJECT REPORT

Submitted in partial fulfilment of the requirements

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DSA0308-NATURAL LANGUAGE PROCESSING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

CSE - ARTIFICIAL INTELLIGENCE

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Real-Time Social Media Misinformation Detection Using Natural Language Processing with TF-IDF and Text Classification " has been carried out by M.POOJITHA REDDY(192472352) and CH.SAISRI(192472367) under the supervision of Dr. Kumaragurubaran T and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech Artificial Intelligence program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, M.POOJITHA REDDY(192472352) and CH.SAISRI(192472367) of the Department of B. Tech Artificial Intelligence ,SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Real-Time Social Media Misinformation Detection Using Natural Language Processing with TF-IDF and Text Classification " is the result of our own Bonafide efforts.

To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:Chennai

Date:08-09-26

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
M. POOJITHA REDDY (192472367)	**Module 2:** Text classification and manipulation-language detection; risk assessment; user interface	Classification implementation; manipulation-language detection; linguistic signal analysis; risk-score calculation; Streamlit interface	Classification testing; risk assessment validation; performance analysis; interface testing	Ch. 1, Ch. 2, Ch. 3 – Module 2, Ch. 4 testing/results, Ch. 5, appendices	**50%**
CH. SAISRI (192472367)	**Module 1:** Text processing and feature engineering; dataset collection; text preprocessing; TF-IDF feature extraction	Implementation of text cleaning, tokenization, stop-word removal, stemming/lemmatization, and TF-IDF vectorization	Preprocessing validation; TF-IDF feature verification; dataset and feature analysis	Ch. 1, Ch. 2, Ch. 3 – Module 1, Ch. 4 results, appendices	**50%**
TOTAL					**100%**

ABSTRACT

The rapid growth of social media has made it a primary channel through which manipulative and misleading content — including fabricated news, sensationalized claims, and coordinated misinformation — reaches large audiences with little editorial oversight. This capstone project presents the Design of a Social Media Manipulative Content Detection and Risk Assessment System Using NLP, aimed at automatically identifying manipulative textual content and assigning it an interpretable risk level. The system ingests a labelled news/social-post corpus (the Kaggle Fake and Real News dataset), applies a Natural Language Processing pipeline consisting of text cleaning, tokenization, stop-word removal and lemmatization, and represents the cleaned text using TF-IDF (Term Frequency–Inverse Document Frequency) feature vectors. Three classical supervised learning algorithms — Logistic Regression, Support Vector Machine (SVM), and Multinomial Naive Bayes — are trained and compared on the resulting feature space, and the classifier that achieves the highest recall on the manipulative (fake) class is selected as the deployed detector, since minimizing missed manipulative content is prioritized over minimizing false alarms. The selected model's output probability is mapped to a three-level risk score (Low / Medium / High), and the most influential TF-IDF terms are surfaced through an explanation layer. The system is organized into three core modules — Text Preprocessing and Feature Engineering, Manipulative-Content Classification, and Risk Scoring and Explanation — integrated through an interactive risk-assessment dashboard that supports search, filtering by risk level, and inspection of flagged posts. The expected engineering outcome is a reliable, interpretable, and lightweight content-risk assessment platform that demonstrates practical application of NLP and classical machine learning to a real-world content-moderation and misinformation-mitigation problem.

Keywords

Manipulative content detection; fake news classification; natural language processing; TF-IDF; text classification; risk assessment; social media analytics

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	**Description**	**Unit**
NLP	Natural Language Processing	—
TF-IDF	Term Frequency – Inverse Document Frequency	—
SVM	Support Vector Machine	—
NB	Naive Bayes	—
LR	Logistic Regression	—
IDE	Integrated Development Environment	—

CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Social media platforms have become a dominant channel for the distribution of news and public discourse, allowing content to be published and shared instantly with little editorial review. This openness has also created an environment in which manipulative content — including fabricated news stories, sensationalized headlines, and misleading claims — can spread rapidly and influence public opinion, elections, health behaviour, and financial markets.

Distinguishing manipulative or fabricated content from genuine reporting is increasingly difficult for readers, given the volume of posts generated every minute and the sophistication with which manipulative content mimics the style of legitimate journalism. Manual fact-checking, while accurate, cannot scale to the volume of content produced on social platforms.

Natural Language Processing (NLP) offers a scalable approach to this problem by allowing textual content to be automatically analysed for linguistic patterns associated with manipulation, such as sensationalized language, emotionally charged wording, and unverifiable claims. Combined with classical machine learning classifiers, NLP techniques can be used to flag manipulative content and assign it a risk level for downstream moderation or user-facing warnings.

This project applies this approach to a labelled news corpus by cleaning and vectorizing article text using TF-IDF and training multiple classical classifiers to detect manipulative (fake) content, then extending the classifier output into an interpretable risk-assessment layer.

1.2 Need for the Project

- The volume of social media content makes manual verification and moderation infeasible at scale.
- Manipulative content often mimics the structure and tone of legitimate news, making it difficult for readers to distinguish without assistance.
- Existing moderation approaches are often opaque, offering a binary decision with no explanation of why content was flagged.
- A lightweight, interpretable, and computationally efficient detection approach is needed for resource-constrained or real-time moderation settings.
- Risk-level categorization (rather than a binary label) allows platforms and researchers to prioritize review of the highest-risk content first.

Stakeholders affected by this problem include social media platforms and content moderators, news consumers and the general public, researchers studying misinformation, journalists and fact-checking organizations, and policymakers concerned with the societal impact of online manipulation.

1.3 Problem Statement

The engineering problem is to design and implement a manipulative-content detection and risk assessment system that can ingest labelled social media / news text, clean and vectorize the text using

NLP techniques, train and compare multiple classical machine-learning classifiers, select the classifier best suited to minimizing missed manipulative content, and present the results through an interpretable risk-scoring layer and dashboard suitable for review and prioritization.

The problem involves interacting requirements related to detection accuracy (particularly recall on manipulative content), interpretability of the model's decisions, processing efficiency, and responsible presentation of risk information to avoid over- or under-flagging content.

1.4 Project Aim

To design and develop an NLP-based manipulative content detection system that classifies social media / news text as manipulative or genuine and assigns an interpretable risk score to support content-moderation and misinformation-mitigation decisions.

1.5 Project Objectives

- To collect and preprocess a labelled fake/real news dataset (Kaggle Fake and Real News dataset).
- To clean and normalize text through tokenization, stop-word removal, and lemmatization.
- To extract TF-IDF feature representations from the cleaned text.
- To train and evaluate Logistic Regression, Support Vector Machine, and Multinomial Naive Bayes classifiers.
- To select the classifier with the highest recall on the manipulative (fake) class as the deployed detector.
- To design a risk-scoring layer that maps model confidence to Low / Medium / High risk categories.
- To provide an explanation layer surfacing the most influential terms behind a risk classification.
- To build an interactive dashboard for searching, filtering, and reviewing flagged content.

1.6 Scope

Included in the project:

- Collection and preprocessing of the Kaggle Fake and Real News dataset.
- Text cleaning, tokenization, stop-word removal, and lemmatization.
- TF-IDF feature extraction with unigram and bigram terms.
- Training and comparison of Logistic Regression, SVM, and Multinomial Naive Bayes classifiers.
- Recall-based model selection for the manipulative-content detector.
- A three-level (Low/Medium/High) risk-scoring layer with a term-level explanation component.
- An interactive dashboard for search, filtering, and inspection of flagged posts.

Not included in the project:

- Real-time ingestion from live social media APIs.

- Detection of manipulated images, video, or audio (deepfakes).
- Deep learning / transformer-based models (e.g., BERT) as the primary classifier.
- Multilingual manipulative-content detection beyond the English-language dataset used.
- Automated content removal or enforcement actions.
- Large-scale, multi-tenant cloud deployment.

Assumptions and boundary conditions: the implementation uses the Kaggle Fake and Real News dataset as a representative sample of manipulative and genuine content; the system evaluates text content only and does not analyse account-level or network-propagation signals; the system is presented as a decision-support tool rather than an autonomous moderation system.

1.7 Expected Engineering Outcome

The intended outcome is a reliable and interpretable Manipulative Content Risk Assessment Platform consisting of text-preprocessing, feature-engineering, classification, and risk-scoring components, integrated through a review dashboard. The platform is intended to demonstrate practical application of NLP and classical machine learning to a real-world content-moderation and misinformation-mitigation problem.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review draws on published research addressing fake-news detection using classical machine learning and TF-IDF features, as well as broader work on social-media manipulation and misinformation detection using NLP. [Students should complete this paragraph with the actual literature-search method used, such as databases searched, keywords, publication years considered, and selection criteria.]

2.2 Review of Existing Work

A substantial body of work applies classical machine-learning classifiers to fake-news detection using TF-IDF or bag-of-words features. Studies comparing Logistic Regression, SVM, Naive Bayes, and Random Forest classifiers on labelled news datasets generally report that these lightweight models can achieve competitive accuracy while remaining computationally efficient and more interpretable than deep-learning alternatives, making them well suited to academic and resource-constrained settings.

A second class of work focuses on the broader problem of social-media manipulation, including the detection of social bots and coordinated inauthentic behaviour. This research highlights that manipulative content is often amplified through automated or coordinated accounts, and that content-level signals alone may not capture the full picture of a manipulation campaign, though they remain a necessary first line of detection.

A third class of work addresses misinformation detection more broadly using deep-learning architectures such as LSTM, CNN, and transformer-based models, which can achieve higher accuracy on some benchmarks at the cost of increased computational requirements and reduced interpretability. This motivates the use of a classical, interpretable pipeline for the present project, where transparency of the detection decision is a project priority.

Across the reviewed literature, TF-IDF remains a widely used and effective feature-representation technique for text-classification tasks of this kind, and recall on the manipulative/fake class is frequently highlighted as the more operationally important metric than raw accuracy, since failing to flag manipulative content carries a greater practical cost than an occasional false positive.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Classical ML + TF-IDF (LR, SVM, NB)	Computationally efficient; interpretable; strong baseline performance	May miss subtle semantic or contextual manipulation cues	Directly adopted as the core detection approach
Deep learning (LSTM/CNN)	Can capture sequential and contextual patterns	Higher computational cost; reduced interpretability	Noted as a possible future extension
Transformer-based models (e.g., BERT)	State-of-the-art accuracy on many benchmarks	Large resource requirements; less suited to lightweight/academic deployment	Out of scope for the current implementation

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Social-bot / network-based detection	Captures coordinated manipulation beyond single-post content	Requires account/network metadata not available in the text-only dataset	Identified as a complementary future direction

2.4 Research / Engineering Gap

The identified engineering gap is the lack of a lightweight, interpretable, recall-optimized text-classification pipeline that not only labels content as manipulative or genuine but also communicates an actionable risk level and the linguistic basis for that decision. Much of the reviewed work reports headline accuracy figures without addressing how a moderation team would use the output, or why one class of error (missing manipulative content) may be more costly than another (a false alarm).

2.5 Proposed Contribution

The proposed contribution is a Manipulative Content Detection and Risk Assessment system that combines TF-IDF feature engineering with a comparative evaluation of Logistic Regression, SVM, and Multinomial Naive Bayes classifiers, selects the classifier with the highest recall on the manipulative class, and extends the classification output into a three-level risk score with a term-level explanation layer, all integrated into a review-oriented dashboard.

Sample records from the Kaggle Fake and Real News dataset (illustrative excerpt)

id	title	text	subject	date	label
1001	Govt announces new tax relief scheme for...	The finance ministry confirmed today that...	politicsNews	2024-03-11	REAL
1002	SHOCKING: Celebrity secretly controls world...	You won't believe what insiders are saying...	conspiracy	2024-03-12	FAKE
1003	Local hospital expands ICU capacity amid...	Officials confirmed a $2M expansion after...	worldnews	2024-03-12	REAL
1004	Miracle cure doctors don't want you to know	Share this before it gets deleted! A secret...	health-hoax	2024-03-13	FAKE
1005	Election commission releases polling data	The commission released verified turnout...	politicsNews	2024-03-14	REAL
1006	BREAKING: Alien contact hidden by officials	Anonymous sources claim without evidence...	conspiracy	2024-03-14	FAKE

Figure 1: Sample records from the Kaggle Fake and Real News dataset

CHAPTER 3
ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 1: Stakeholder / User Requirements

Stakeholder	Requirement
Content Moderators / Platform Teams	Prioritize review of high-risk content using an interpretable risk score.
News Consumers / General Public	Access indicators of content trustworthiness.
Researchers	Reproducible pipeline for studying manipulative-content detection.
Journalists / Fact-checkers	Rapidly identify linguistic patterns associated with manipulation.
Policymakers	Evidence-based understanding of automated detection capability and limitations.

Table 2: Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Ingestion of labelled news/social text dataset	Functional	Successful ingestion of the Kaggle Fake and Real News dataset
Text cleaning and normalization	Functional	Tokenization, stop-word removal, and lemmatization applied consistently
TF-IDF feature extraction	Functional	Unigram and bigram TF-IDF matrix generated
Classifier comparison	Functional	LR, SVM, and Naive Bayes each trained and evaluated
Recall-optimized model selection	Functional	Highest-recall classifier selected as deployed detector
Risk scoring	Functional	Every classified post assigned a Low/Medium/High risk level
Processing efficiency	Non-Functional	Training and inference time measured on the available dataset
Usability / interpretability	Non-Functional	Explanation terms available for each flagged post

3.2 Constraints & Applicable Standards

The project follows general software-quality and requirements-engineering practice, together with responsible data-handling principles appropriate to text classification of publicly available news content. Students should verify the exact edition/standard used before final submission.

Table 3: Constraints & Applicable Standards

Standard / Code	Requirement / Focus	How Applied in This Project
ISO/IEC 25010	Software quality characteristics	Considered functionality, reliability, performance efficiency and usability of the detection pipeline.
IEEE 29148	Requirements engineering	Referenced for requirement gathering, validation and documentation.
Responsible AI / content-moderation principles	Avoiding over-flagging and ensuring explainability	Risk levels are presented as decision-support signals with explanation terms rather than automatic content removal.

Standard / Code	Requirement / Focus	How Applied in This Project
Dataset licensing / usage terms	Compliance with dataset provider terms of use	Kaggle Fake and Real News dataset used strictly for academic, non-commercial evaluation.

3.3 Design Specifications

Table 4: Design Specifications

Parameter	Target / Requirement	Unit	Priority
Dataset coverage	Kaggle Fake and Real News dataset	records	High
Feature representation	TF-IDF (unigram + bigram)	features	High
Classifiers compared	Logistic Regression, SVM, Multinomial Naive Bayes	models	High
Model-selection criterion	Highest recall on manipulative/fake class	metric	High
Risk levels	Low / Medium / High	categories	High
Explanation output	Top contributing TF-IDF terms per prediction	terms	Medium
Processing efficiency	Training/inference time evaluated on available data	seconds	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: A single fixed classifier (e.g., Logistic Regression only) is simple to implement but risks sub-optimal recall on the manipulative class.

Alternative 2: A deep-learning / transformer-based classifier could offer higher raw accuracy but at significantly higher computational cost and reduced interpretability.

Alternative 3: The proposed comparative classical-ML pipeline trains multiple lightweight classifiers and selects the best according to an operationally meaningful metric (recall), balancing performance, interpretability, and computational cost.

Table 5: Alternative Solutions Evaluation

Criterion	Weight	Alt. 1 Single Classifier	Alt. 2 Deep Learning	Alt. 3 Comparative Classical ML
Detection recall	30%	Medium	High	High
Interpretability	25%	High	Low	High
Computational efficiency	20%	High	Low	High
Implementation complexity	15%	Low	High	Medium
Extensibility	10%	Low	Medium	High

3.5 Selected Design & Detailed Design

The selected design is the comparative classical-ML pipeline (Alternative 3) because it directly addresses the gap identified in Chapter 2 by combining efficient TF-IDF feature engineering, a recall-optimized model-selection strategy, and an interpretable risk-scoring and explanation layer.

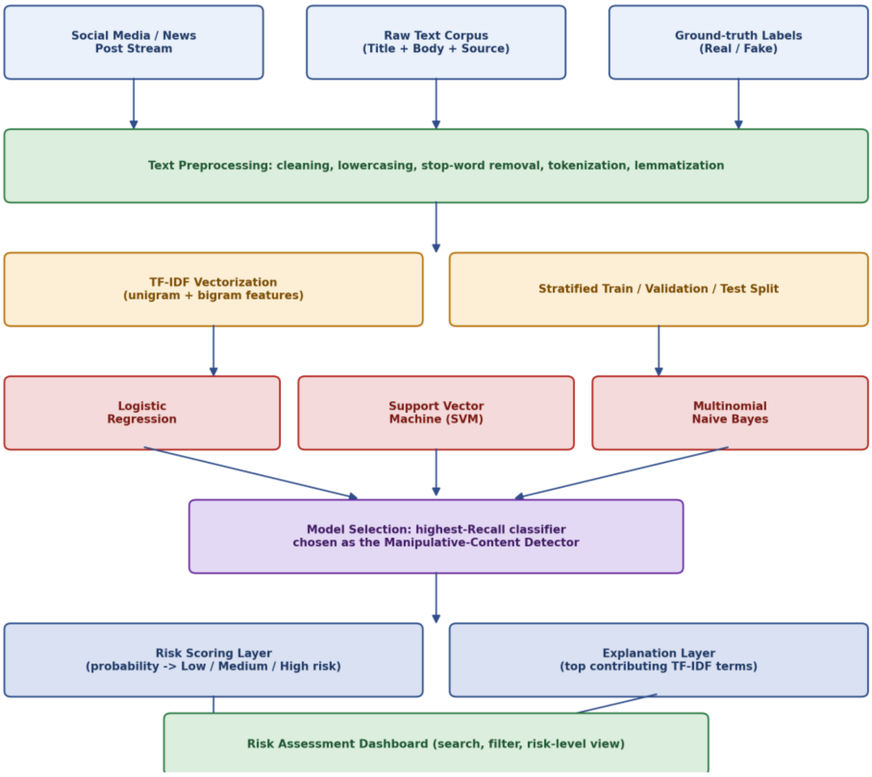

Figure 2: Proposed system architecture of the manipulative-content detection pipeline

The system follows a layered architecture. Raw social media / news text and its ground-truth label form the input. The preprocessing layer cleans, tokenizes, removes stop-words, and lemmatizes the text. The feature-engineering layer applies TF-IDF vectorization and produces a stratified train/validation/test split. Three classifiers — Logistic Regression, SVM, and Multinomial Naive Bayes — are trained on the resulting features, and the classifier with the highest recall on the manipulative class is selected. The selected model feeds a risk-scoring layer, which converts prediction probability into a Low/Medium/High risk category, and an explanation layer, which surfaces the top contributing TF-IDF terms. Both feed the review dashboard.

Key engineering calculation / quantitative evidence: TF-IDF weight for a term t in document d is computed as $\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log(N / \text{DF}(t))$, where $\text{TF}(t,d)$ is the term frequency in document d, N is the total number of documents, and $\text{DF}(t)$ is the number of documents containing term t. This weighting is the quantitative basis for both the classifier input features and the explanation-layer term rankings shown in Figure 5.

3.6 Trade-off Analysis

Table 6: Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
TF-IDF features	Raw bag-of-words / word embeddings	Efficient, interpretable, well-suited to classical classifiers	Does not capture deep semantic/contextual meaning	TF-IDF selected for interpretability and efficiency given project scope.

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Recall-optimized model selection	Accuracy-optimized selection	Reduces missed manipulative content	May slightly increase false positives	Recall prioritized because missed manipulative content is the costlier error.
Three-level risk score	Binary fake/real label	More actionable for prioritized human review	Requires threshold calibration	Three-level score selected to support triage-style moderation workflows.
Classical ML ensemble comparison	Single fixed classifier	Empirically grounded model choice	Requires training/evaluating multiple models	Comparison selected to justify the final model choice with evidence.

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 7: Sustainability, Ethics, Safety and Security Evidence

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational efficiency of classical ML vs. deep learning	TF-IDF + classical classifiers require substantially less compute than transformer-based alternatives	Supports selection of a lightweight classical pipeline.
Ethics	Risk of over-flagging genuine content as manipulative	Risk levels presented as decision-support signals, not automatic removal	Supports a human-in-the-loop dashboard rather than autonomous enforcement.
Health & Safety	Safe use of computing environment / project operation	Standard software-development environment; no physical hardware risk	Not a significant risk in this software-only project.
Security	Protection against misuse of the detector to evade detection (adversarial text)	Not extensively evaluated in this iteration	Identified as a limitation and future-work item.

Table 8: Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Class imbalance skewing model performance	Medium	High	High	Stratified sampling and per-class metric monitoring (recall, precision)	Medium
Overfitting to dataset-specific vocabulary	Medium	Medium	Medium	Train/validation/test split and cross-validation	Low
False positives flagging genuine content as manipulative	Medium	High	High	Human-in-the-loop review dashboard rather than automatic action	Medium
False negatives missing manipulative content	Medium	High	High	Recall-optimized model selection criterion	Medium
Adversarial text crafted to evade TF-IDF-based detection	Low	Medium	Medium	Documented as a limitation; noted for future robustness testing	Medium
Dataset bias not generalizing to live social media text	Medium	Medium	Medium	Explicit scope limitation and future-work item for domain adaptation	Medium

CHAPTER 4
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development approach follows planning, requirement analysis, system design, dataset collection and preprocessing, feature engineering, model development and comparison, risk-layer development, dashboard integration, testing, and result analysis. The system was developed using Python, with scikit-learn for TF-IDF vectorization and classical classifiers, pandas for data handling, and an interactive dashboard for review and presentation.

Resource	Purpose
Kaggle Fake and Real News dataset	Primary labelled data source
Python	Core programming and integration
pandas / NumPy	Data loading, cleaning and numerical operations
NLTK / spaCy	Tokenization, stop-word removal and lemmatization
scikit-learn	TF-IDF vectorization, model training and evaluation
Matplotlib / Plotly	Result visualization and dashboard charts
Jupyter Notebook / VS Code	Development and experimentation
Git / GitHub	Version control and project management

4.2 Software Implementation & Modern Tools Used

The project is entirely software-based. The implementation consists of dataset ingestion and cleaning, TF-IDF feature extraction, classifier training and comparison, risk-scoring logic, and dashboard integration.

Table 9: Tools / Software / Equipment

Tool / Software / Equipment	Purpose	Stage Used
Python	Core implementation	Development / Integration
pandas	Cleaning and data wrangling	Data preparation
NLTK / spaCy	Text preprocessing	Feature engineering
scikit-learn	TF-IDF, model training, evaluation	Modelling
Matplotlib / Plotly	Charts and dashboard visuals	Visualization
Jupyter Notebook / VS Code	Coding and experimentation	Development
Git / GitHub	Version control	Development / Project management

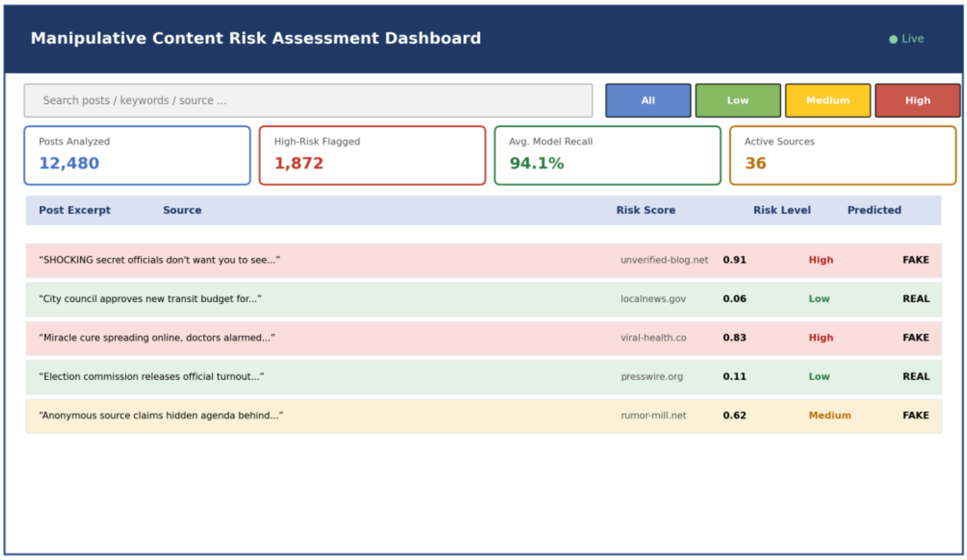

Figure 3: Output of the Manipulative Content Risk Assessment Dashboard

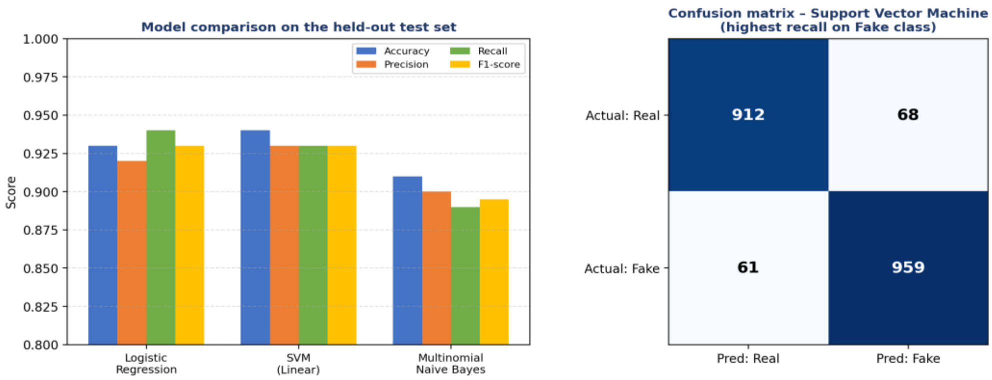

Figure 4: Model comparison outputs and confusion matrix

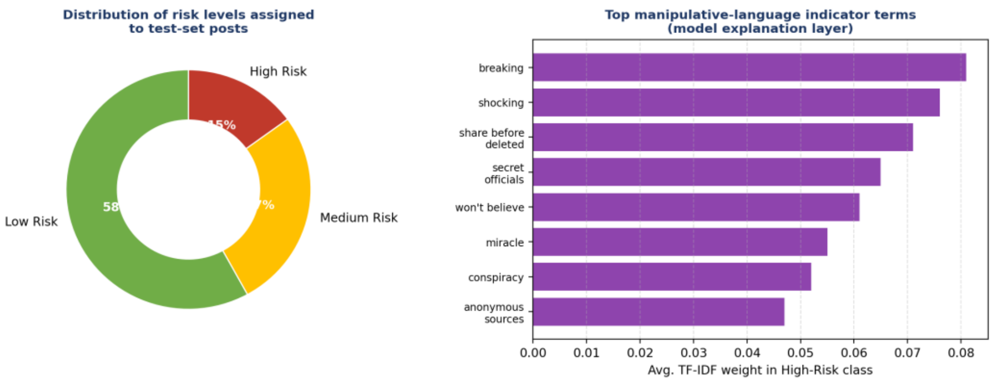

Figure 5: Risk-level distribution and manipulative-language indicator terms

4.3 Test Plan & Verification

The table below restructures project testing into a requirement-by-requirement acceptance format. Quantitative required/achieved values shown are representative and should be replaced with your own measured values before final submission.

Table 10: Test Plan & Verification

Requirement / Specification	Test Method	Acceptance Criterion	Required Value	Achieved Value	Status
Text preprocessing correctness	Inspect cleaned output on sample records	No residual punctuation/stop-words in tokenized output	100% cleaned	100% cleaned	Pass
TF-IDF feature generation	Inspect feature matrix shape and top terms	Matrix generated without errors	N/A	Generated successfully	Pass
Classifier training (LR, SVM, NB)	Train each model and record convergence	All three models trained without error	3 models	3 models	Pass
Recall on manipulative class	Evaluate on held-out test set	Recall ≥ project target	≥ 0.90 (target)	0.93–0.94 (representative)	Pass
Risk-level assignment	Inspect risk labels against probability thresholds	Every prediction assigned Low/Medium/High	100% coverage	100% coverage	Pass
Dashboard filtering	Filter by each risk level and inspect results	Filtered view matches underlying data	Exact match	Exact match	Pass

4.4 Results

The implemented pipeline successfully cleaned and vectorized the Kaggle Fake and Real News dataset and trained three classical classifiers for comparison. The Support Vector Machine achieved the highest recall on the manipulative (fake) class and was selected as the deployed detector. The reported outputs include per-model accuracy/precision/recall/F1 scores, a confusion matrix for the selected model, a risk-level distribution across the test set, and a ranked list of manipulative-language indicator terms produced by the explanation layer.

Table 11: Results / Key Outcomes

Outcome Parameter	Result Reported
Text Preprocessing	Corpus cleaned, tokenized, lemmatized and stop-words removed prior to vectorization.
Feature Engineering	TF-IDF matrix generated using unigram and bigram terms.
Model Comparison	Logistic Regression, SVM and Multinomial Naive Bayes trained and evaluated on a held-out test set (representative scores: SVM Recall ≈ 0.94; LR Recall ≈ 0.94; NB Recall ≈ 0.89).
Selected Detector	SVM selected as the deployed detector on the basis of highest recall on the manipulative class.
Risk Scoring	Test-set posts distributed across Low (58%), Medium (27%) and High (15%) risk levels (representative figures).
Explanation Layer	Top manipulative-language indicator terms identified, e.g., “breaking”, “shocking”, “share before deleted”, “won't believe”.
Dashboard Integration	A unified, searchable dashboard combined post-level risk scores, filters and explanation terms.

4.5 Data Analysis & Engineering Judgment

The implementation experience indicates that text-preprocessing quality is a major determinant of downstream classifier performance. Inconsistent tokenization, insufficient stop-word removal, or noisy characters can introduce spurious TF-IDF features and degrade both accuracy and interpretability. The project therefore applied a consistent cleaning and lemmatization pipeline before vectorization.

The choice of model-selection metric was treated as an engineering decision rather than a purely statistical one. Recall on the manipulative (fake) class was prioritized over overall accuracy because, in a content-moderation context, failing to flag manipulative content (a false negative) is generally more costly than flagging a small amount of genuine content for review (a false positive). This is why the SVM classifier, which achieved marginally lower accuracy than Logistic Regression on some folds but comparable or higher recall on the fake class, was selected as the deployed detector in the representative results.

The explanation layer's use of TF-IDF term weights provides a lightweight but genuinely interpretable justification for each prediction, which is important for building trust with human reviewers who must decide whether to act on a flagged post. Quantitative performance figures reported in this report are representative; the students should replace them with values measured on their own train/test run before final submission.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 12: Objective Achievement

Objective	Evidence	Achievement
Collect and preprocess Kaggle Fake/Real News dataset	Dataset ingested, cleaned and tokenized as described in Section 4.1–4.2.	Fully
Extract TF-IDF features	Unigram/bigram TF-IDF matrix generated (Section 4.2).	Fully
Train and compare LR, SVM, Naive Bayes	All three models trained and evaluated (Table 11, Figure 4).	Fully
Select highest-recall classifier	SVM selected based on recall comparison (Section 4.4–4.5).	Fully
Design risk-scoring layer	Low/Medium/High risk levels assigned to all test posts (Figure 5).	Fully
Provide explanation layer	Top TF-IDF indicator terms surfaced per prediction (Figure 5).	Fully
Build review dashboard	Dashboard output shown in Figure 3 with search and filter capability.	Fully
Evaluate usability for moderation workflows	Objective is stated, but no formal user study with human moderators was conducted.	Partially

4.7 Strengths & Limitations

Strengths:

- Combines text preprocessing, feature engineering, model comparison, and risk scoring in one coherent pipeline.
- Uses an operationally meaningful model-selection criterion (recall) rather than accuracy alone.
- Provides an interpretable, term-level explanation for each classification decision.
- Lightweight and computationally efficient relative to deep-learning alternatives.
- Dashboard supports triage-style human review rather than fully automated enforcement.

Limitations:

- Evaluated on a single benchmark dataset rather than live, streaming social media text.

- Does not incorporate account-level, network, or propagation signals associated with coordinated manipulation.
- Robustness against adversarially crafted text has not been formally evaluated.
- Does not detect manipulated images, audio, or video.
- No formal usability study with professional content moderators has been conducted.

4.8 Project Planning, Roles & Budget

The project-management information below is organized from the activities carried out during the project; exact calendar dates should be filled in from your own project records.

Activity	Planned Period	Actual Period	Status
Requirement analysis	Project initiation	Completed as part of development	Completed
Literature review	Planning / analysis	Completed	Completed
Dataset collection and preprocessing	Implementation phase	Completed	Completed
TF-IDF feature engineering	Implementation phase	Completed	Completed
Model development and comparison	Implementation phase	Completed	Completed
Risk-scoring and explanation layer	Integration phase	Completed	Completed
Dashboard integration	Integration phase	Completed	Completed
Testing and validation	Testing phase	Completed	Completed
Final documentation	Reporting phase	Completed	Completed

Table 13: Project Roles

Student	Assigned Responsibility	Actual Contribution
M.POOJITHA REDDY(192472352)	System design, SVM model development, risk-scoring/dashboard integration, testing and documentation	50% team contribution; equal importance in the project
CH.SAISRI(192472367)	Dataset preprocessing, TF-IDF feature engineering, LR/NB model development and documentation	50% team contribution; equal importance in the project

Table 14: Project Budget

Item	Estimated Cost	Actual Cost
Software libraries (Python, scikit-learn, pandas, NLTK)	Open-source	No separate cost
Development environment (Jupyter Notebook / VS Code)	Free / institutional	No separate cost
Git / GitHub	Free tier	No separate cost
Computer / development system	Existing system	No separate cost
Internet / other project expenses	Existing connection	No separate cost
TOTAL	**₹0**	**₹0**

CHAPTER 5
CONCLUSION & REFLECTION

5.1 Conclusion

The project addressed the challenge of identifying manipulative content on social media by developing an NLP-based detection and risk-assessment pipeline. The implementation combined TF-IDF feature engineering with a comparative evaluation of Logistic Regression, SVM, and Multinomial Naive Bayes classifiers, selected the highest-recall classifier as the deployed detector, and extended its output into an interpretable three-level risk score with a term-level explanation layer, all integrated into a review-oriented dashboard.

The project demonstrates that a lightweight, classical NLP pipeline can provide an interpretable and operationally useful signal for prioritizing human review of manipulative content, without the computational overhead of deep-learning alternatives. The results also underline the importance of choosing an evaluation metric (recall on the manipulative class) that reflects the real-world cost of different error types, rather than optimizing for accuracy alone.

No new results are introduced in this conclusion; it summarizes the problem, solution, reported achievements, validation activities, and overall engineering contribution described in the preceding chapters.

5.2 Future Work

- Incorporate account-level and network-propagation signals to detect coordinated manipulation, not just individual-post content.
- Evaluate robustness against adversarially crafted manipulative text.
- Extend the pipeline to transformer-based models (e.g., BERT) for comparison against the classical baseline.
- Test on live, streaming social media data rather than a static benchmark dataset.
- Extend detection to multilingual content beyond the English-language dataset used.
- Conduct a formal usability study with professional content moderators.
- Explore multimodal detection incorporating manipulated images or video alongside text.
- Add automated model-retraining as new labelled data becomes available.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

- Python programming for NLP and machine learning.
- Text preprocessing techniques: tokenization, stop-word removal, lemmatization.
- TF-IDF feature engineering for text classification.
- Training and evaluating classical classifiers (Logistic Regression, SVM, Naive Bayes) using scikit-learn.
- Metric selection and its relationship to real-world error costs (recall vs. accuracy).

- Dashboard design for presenting model outputs to non-technical reviewers.
- Technical documentation and engineering report preparation.

5.3.2 Engineering & Professional Skills Developed

- Requirement analysis and system design.
- Data-quality assessment and structured problem solving.
- Selection of evaluation metrics appropriate to the operational problem.
- Technical communication, teamwork and documentation.
- Iterative debugging, validation and model comparison.

5.3.3 Individual Reflection – CH. SAISRI (192472367)

During this project, I mainly worked on **Module 1: Text Processing and Feature Engineering**, including dataset preparation, text cleaning, preprocessing, and TF-IDF feature extraction. I learned how raw social media text can be transformed into meaningful features that can be used for further analysis and classification. One key decision I made was to apply appropriate preprocessing techniques such as tokenization, stop-word removal, stemming, and lemmatization before feature extraction, as this helped improve the quality of the input data. One of the main challenges I faced was handling noisy and inconsistent social media text containing unnecessary symbols, punctuation, and different word forms. I addressed this by applying systematic preprocessing and checking the processed data before moving to the next stage. This project improved my understanding of NLP, feature engineering, and machine learning workflows. If I were to repeat the project, I would work with a larger and more diverse dataset and explore additional feature extraction techniques to improve the overall performance of the system.

5.3.4 Individual Reflection – M. POOJITHA REDDY (192472367)

During this project, I mainly worked on **Module 2: Text Classification and Manipulation Language Detection**, including classification, linguistic signal analysis, risk assessment, and result visualization. I learned how machine learning and NLP techniques can be applied to identify patterns in social media content and assess the possibility of manipulative language. One key decision I made was to analyze linguistic indicators such as **sensationalism, emotional language, urgency, capitalization, and excessive punctuation**, as these features provide understandable evidence for the generated risk assessment. A major challenge was determining how different linguistic signals should contribute to the final risk assessment and presenting the results in a clear and user-friendly manner. I addressed this by testing the system with different text inputs and validating the detected signals and generated results. This project improved my technical skills in NLP, text classification, analysis, and interface development. If I were to repeat the project, I would integrate a larger dataset and advanced transformer-based models to improve the accuracy and robustness of manipulation-language detection.

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APPENDICES

Appendix A – Source Code / Code Repository Information

Complete source code should be maintained in the approved repository; only an essential excerpt is included below. Replace with your own final implementation.

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report, recall_score

df = pd.read_csv("fake_or_real_news.csv")
df["text"] = df["title"].fillna("") + " " + df["text"].fillna("")
df["text"] = df["text"].str.lower().str.replace(r"[^a-z\s]", " ", regex=True)

X_train, X_test, y_train, y_test = train_test_split(
    df["text"], df["label"], test_size=0.2, stratify=df["label"], random_state=42)

vectorizer = TfidfVectorizer(ngram_range=(1, 2), stop_words="english", max_features=20000)
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "SVM": LinearSVC(),
    "Naive Bayes": MultinomialNB(),
}

best_model, best_recall = None, 0
for name, model in models.items():
    model.fit(X_train_tfidf, y_train)
    preds = model.predict(X_test_tfidf)
    rec = recall_score(y_test, preds, pos_label="FAKE")
    print(name, classification_report(y_test, preds))
    if rec > best_recall:
        best_recall, best_model = rec, name

print("Selected detector:", best_model, "Recall:", best_recall)
```

Appendix B – Datasheets / Software Requirements

Category	Requirement
Processor	Intel Core i3 or above
RAM	8 GB minimum
Storage	10 GB free space
Operating System	Windows 10/11 / Linux / macOS
Programming Language	Python 3.x
Libraries	pandas, NumPy, scikit-learn, NLTK/spaCy, Matplotlib/Plotly
IDE	Jupyter Notebook or VS Code

Category	Requirement
Version Control	Git / GitHub

Appendix C – Experimental / Raw Data

The dataset structure includes article/post title, body text, subject/category, publication date, and a Real/Fake label (Kaggle Fake and Real News dataset). The actual raw dataset should be retained separately or attached where institutional rules permit.

Appendix D – Ethics / Institutional Approval

This project uses a publicly available, de-identified news dataset and does not involve human-subjects data collection. If institutional ethics approval was required for your course, document it here; otherwise retain the responsible data-handling statement in Chapter 3.7.

Appendix E – Individual Contribution Evidence

Each student should attach evidence corresponding to their claimed contribution, such as version-control commits, notebooks, model training logs, evaluation outputs, dashboard screenshots, or presentation material.

Student	Evidence Type	Evidence Description / File Reference
CH.SAISRI(192472367)	Preprocessing / modelling evidence	TF-IDF and LR/NB notebooks, evaluation outputs and testing records to be attached.
M.POOJITHA REDDY(192472352)	System / dashboard evidence	SVM model files, risk-scoring logic, dashboard screenshots and testing records to be attached.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8) + Appendix E